

Electronic Companion

EC.1. Notation and Symbols

Table EC.1 Abbreviations Used in the Main Manuscript and the Online Supplement

Abbreviation	Meaning	Abbreviation	Meaning
3-SRC	3-subset-row cut	AFF	Arc-flow formulation
BB	Branch-and-bound	BF	Best-fit algorithm
BF1	Improved best-fit algorithm	BF2	Bidirectional best-fit algorithm
BP	Branch-and-price	BPC	Branch-price-and-cut algorithm
BPP	Bin packing problem	BPPC	Bin packing problem with conflicts
CG	Column generation	CPU	Central processing unit
CRG	Column-and-row generation	CSP	Cutting stock problem
DFS	Depth-first search	DH	Diving heuristic
DP	Dynamic programming	ET	Early termination
GB	Gigabyte	GP	Grouping procedure
LB	Lower bound	LCF	Least-compatible-first heuristic
LMP	Linear master problem	LNSA	Large-neighborhood search algorithm
LP	Linear programming	LSA	Label-setting algorithm
MP	Master problem	PE	Pattern enumeration
pp	Percentage points	PSP	Pricing subproblem
RF	Ryan-Foster	RH	Rounding heuristic
RLMP	Restricted linear master problem	RSCP	Restricted set-covering problem
SR	Subset-row	UB	Upper bound
VSBBP	Variable-sized bin packing problem	VSBBPC	VSBBP with conflicts
VSBBPTW	VSBBP with time windows		

EC.2. Proofs

EC.2.1. Proof of Proposition 1 (Grouping Exactness)

Proof: Consider an arbitrary search node and bin type $b \in \mathcal{B}$. Let $\bar{\mathcal{G}}$ be the compatibility graph whose edges join pairs of items without an original conflict, and let $\mathcal{Q}$ be the set of its maximal cliques. Let $\mathcal{P}_b^{\text{node}}$ be the set of all capacity-feasible patterns of bin type b that satisfy the original conflicts and all current RF constraints. For each $Q \in \mathcal{Q}$, let $\mathcal{P}_b(Q)$ be the feasible-pattern set of the grouped pricing problem over Q before the dual-based reduction. In this grouped problem, a current binding component $\mathcal{C}$ is available only if $\mathcal{C} \subseteq Q$ and is selected or rejected as a whole. At the same time, we explicitly enforce every conflict constraint created by branching. Consider any pattern $p \in \mathcal{P}_b^{\text{node}}$. Because p satisfies every original conflict constraint, its item set $\mathcal{I}_p$ is a clique of $\bar{\mathcal{G}}$. Every clique of a finite graph is contained in a maximal clique, so $\mathcal{I}_p \subseteq Q$ for some $Q \in \mathcal{Q}$. Because p also satisfies every current binding constraint, each binding component selected by p is contained in $\mathcal{I}_p$ and hence in Q . The grouped problem over Q therefore admits all components selected by p , and p satisfies its explicitly enforced conflict constraints. Thus, $p \in \mathcal{P}_b(Q)$, which proves $\mathcal{P}_b^{\text{node}} \subseteq \bigcup_{Q \in \mathcal{Q}} \mathcal{P}_b(Q)$. Conversely, consider any pattern $p \in \mathcal{P}_b(Q)$ for some $Q \in \mathcal{Q}$. Because $\mathcal{I}_p \subseteq Q$ and Q is a clique of $\bar{\mathcal{G}}$, p contains no originally conflicting pair. By construction, the grouped problem selects each admitted binding component as a whole and enforces every conflict constraint created by branching. Hence, p satisfies all current RF constraints and belongs to $\mathcal{P}_b^{\text{node}}$. Therefore, $\mathcal{P}_b^{\text{node}} = \bigcup_{Q \in \mathcal{Q}} \mathcal{P}_b(Q)$. The reduced cost of a pattern, including its active SR terms, depends on the pattern and the current dual solution, but not on the maximal clique that generates it. Taking the minimum reduced cost over the two equal feasible-pattern sets therefore gives the same pricing optimum. A pattern contained in several maximal cliques may be generated more than once, but duplicate generation does not affect this minimum. The componentwise argument in Section 4.1 of the main manuscript further shows that the subsequent dual-based reduction preserves the optimum of every grouped pricing problem. Hence, the minimum over all reduced grouped problems equals the minimum obtained by pricing over the complete item set. Q.E.D.

EC.2.2. Proof of Proposition 2 (Label Dominance)

Proof: Following the feasible-continuation argument for Dominance 1 cited in Section 4.1 of the main manuscript, consider any feasible continuation $\mathcal{S}_2 \subseteq \mathcal{F}_a$ of $\mathcal{L}_2$ and construct $\mathcal{S}_1 = \mathcal{S}_2 \setminus \mathcal{U}_1$. Since $\mathcal{S}_2 \cap \mathcal{U}_2 = \emptyset$, every item in $\mathcal{S}_2 \setminus \mathcal{S}_1$ belongs to $(\mathcal{U}_1 \setminus \mathcal{U}_2) \cap \mathcal{F}_a$. A feasible continuation selects each current RF binding component as a whole, and $\mathcal{U}_1$ is closed under these components. Therefore, $\mathcal{S}_2 \setminus \mathcal{S}_1 = \mathcal{S}_2 \cap \mathcal{U}_1$ is a union of complete components, so $\mathcal{S}_1$ preserves all binding constraints. The inclusion $\mathcal{S}_1 \subseteq \mathcal{S}_2$ preserves conflicts within the continuation, and $\mathcal{S}_1 \cap \mathcal{U}_1 = \emptyset$ preserves conflicts between the continuation and the partial pattern of $\mathcal{L}_1$. Its weight is at most that of $\mathcal{S}_2$ and hence at most $r_2 \leq r_1$. Because $\mathcal{P}_{l,1} \neq \emptyset$, appending $\mathcal{S}_1$ yields a nonempty pattern. Thus, $\mathcal{S}_1$ is a feasible continuation of $\mathcal{L}_1$.

For $k \in \{1, 2\}$ and $t \in \mathcal{T}$, let $\chi_{kt} = 1$ if $t \in \mathcal{H}_k$ and $\chi_{kt} = 0$ otherwise, and let $\kappa_{kt} = |\mathcal{S}_k \cap \mathcal{I}_t|$. Because $\mathcal{I}_t$ contains exactly three items, the additional penalty associated with t after appending $\mathcal{S}_k$ is $(-\xi_t) \lfloor (\chi_{kt} + \kappa_{kt})/2 \rfloor$. Hence, the resulting change in reduced cost is $\Gamma_k(\mathcal{S}_k) = -\sum_{i \in \mathcal{S}_k} \mu_i + \sum_{t \in \mathcal{T}} (-\xi_t) \lfloor \frac{\chi_{kt} + \kappa_{kt}}{2} \rfloor$. The item dual value removed from the continuation is $\sum_{i \in \mathcal{S}_2 \setminus \mathcal{S}_1} \mu_i$, which is at most $\sum_{i \in (\mathcal{U}_1 \setminus \mathcal{U}_2) \cap \mathcal{F}_a} \mu_i$ because $\mu_i \geq 0$. Moreover, $\mathcal{S}_1 \subseteq \mathcal{S}_2$ implies $\kappa_{1t} \leq \kappa_{2t}$ for every t . If $\chi_{1t} = \chi_{2t}$ or $(\chi_{1t}, \chi_{2t}) = (0, 1)$, the additional penalty associated with t for $\mathcal{L}_1$ is no greater than that for $\mathcal{L}_2$. If $(\chi_{1t}, \chi_{2t}) = (1, 0)$, it exceeds the latter by at most $-\xi_t$. Since $\xi_t \leq 0$, it follows that

$$\begin{aligned} \bar{\delta}_1 + \Gamma_1(\mathcal{S}_1) - \bar{\delta}_2 - \Gamma_2(\mathcal{S}_2) &\leq \bar{\delta}_1 - \bar{\delta}_2 + \sum_{i \in (\mathcal{U}_1 \setminus \mathcal{U}_2) \cap \mathcal{F}_a} \mu_i + \sum_{t \in \mathcal{H}_1 \setminus \mathcal{H}_2} (-\xi_t) \\ &= \bar{\delta}_1 - \bar{\delta}_2 + g_{12} \leq 0. \end{aligned}$$

Every feasible completion of $\mathcal{L}_2$ therefore has a feasible counterpart from $\mathcal{L}_1$ with no higher reduced cost. Thus, $\mathcal{L}_1$ dominates $\mathcal{L}_2$, and $\mathcal{L}_2$ can be discarded without changing the optimal pricing value. Q.E.D.

EC.2.3. Proof of Proposition 3 (Dual Bounds)

Proof: Recall from Proposition 3 that $C_{\min} = \min_{b \in \mathcal{B}} C_b$, $N^{\text{UB}} = \lfloor UB/C_{\min} \rfloor$, and $\delta_{\min}^* = \min_{b \in \mathcal{B}} \delta_b^*$. Let $\mathbf{x}$ be any feasible solution to the full LMP at the current node. Define

$$N_b(\mathbf{x}) = \sum_{p \in \mathcal{P}_b^{\text{node}}} x_p \quad \text{and} \quad \sigma_t(\mathbf{x}) = \sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}: |\mathcal{I}_p \cap \mathcal{I}_t| \geq 2} x_p.$$

Let μ , $\underline{\eta}$, $\bar{\eta}$, and ξ be an optimal dual solution of the current RLMP. For every pattern $p \in \mathcal{P}_b^{\text{node}}$, including a potential pattern, define

$$\delta_p = C_b - \sum_{i \in \mathcal{I}} a_{ip} \mu_i - \underline{\eta}_b - \bar{\eta}_b - \sum_{t \in \mathcal{T}: |\mathcal{I}_p \cap \mathcal{I}_t| \geq 2} \xi_t.$$

Expanding the objective of $\mathbf{x}$ by this definition gives

$$\begin{aligned} \sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}} C_b x_p &= \sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}} \delta_p x_p + \sum_{i \in \mathcal{I}} \mu_i \sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}} a_{ip} x_p \\ &\quad + \sum_{b \in \mathcal{B}} (\underline{\eta}_b + \bar{\eta}_b) N_b(\mathbf{x}) + \sum_{t \in \mathcal{T}} \xi_t \sigma_t(\mathbf{x}). \end{aligned}$$

By feasibility, every item is covered at least once, $L_b \leq N_b(\mathbf{x}) \leq U_b$, and $\sigma_t(\mathbf{x}) \leq 1$. Combining these inequalities with $\mu_i \geq 0$, $\underline{\eta}_b \geq 0$, $\bar{\eta}_b \leq 0$, and $\xi_t \leq 0$ yields

$$\begin{aligned} \sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}} C_b x_p &\geq \sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}} \delta_p x_p + \sum_{i \in \mathcal{I}} \mu_i + \sum_{b \in \mathcal{B}} (L_b \underline{\eta}_b + U_b \bar{\eta}_b) + \sum_{t \in \mathcal{T}} \xi_t \\ &= z^R + \sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}} \delta_p x_p, \end{aligned}$$

where the equality follows from strong duality for the current RLMP. This argument does not require the current RLMP dual solution to satisfy the constraints associated with potential columns not included here.

Every RLMP column has nonnegative reduced cost. For a fixed bin type b , let $\omega_b(\mathbf{x}) = \sum_{p \in \mathcal{P}_b^{\text{node}} \setminus \bar{\mathcal{P}}_b} x_p$ be the total value of potential type- b patterns. Exact pricing gives $\delta_p \geq \delta_b^*$ for each such pattern. If $\delta_b^* < 0$, then $\omega_b(\mathbf{x}) \leq N_b(\mathbf{x}) \leq U_b$ and $\sum_{p \in \mathcal{P}_b^{\text{node}} \setminus \bar{\mathcal{P}}_b} \delta_p x_p \geq \omega_b(\mathbf{x}) \delta_b^* \geq U_b \delta_b^*$. If $\delta_b^* = 0$, the same reduced-cost sum

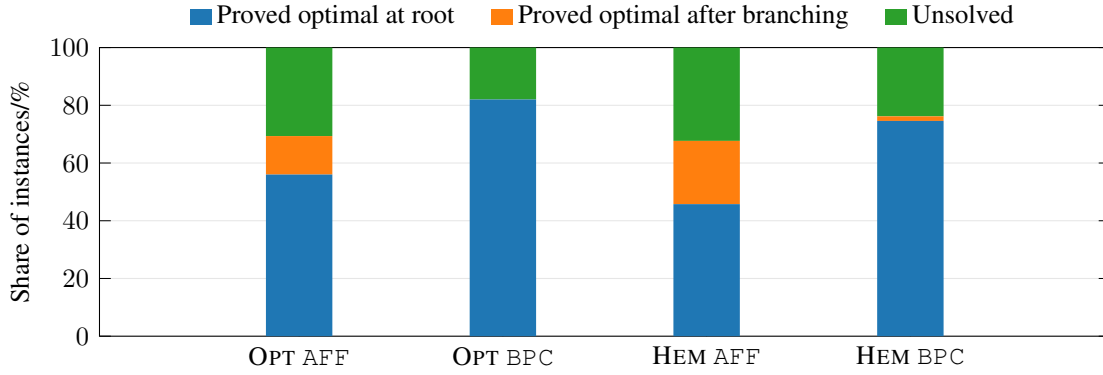


Figure EC.1 Root-Node Optimality and Branching Outcomes on OPT and HEM

is nonnegative. Summing over the bin types proves that every feasible full-LMP solution has objective value at least L^{type} .

Now suppose that $\mathbf{x}$ is a feasible integer solution to the full MP at the current node and improves the incumbent. Its integral cost is at most $UB - 1$. If $N(\mathbf{x}) = \sum_{b \in \mathcal{B}} N_b(\mathbf{x})$ is the number of selected patterns, then $N(\mathbf{x}) \leq \left\lfloor \frac{UB-1}{C_{\min}} \right\rfloor \leq N^{\text{UB}}$. Every selected restricted column has reduced cost at least zero, and every selected potential column of bin type b has reduced cost at least $\delta_b^* \geq \delta_{\min}^*$. Because $\delta_{\min}^* < 0$, $\sum_{b \in \mathcal{B}} \sum_{p \in \mathcal{P}_b^{\text{node}}} \delta_p x_p \geq N(\mathbf{x}) \delta_{\min}^* \geq N^{\text{UB}} \delta_{\min}^*$. Substituting this inequality into the reduced-cost decomposition proves that every feasible integer solution whose objective value is smaller than UB has an objective value of at least L^{inc} , which completes the proof of Proposition 3.

EC.3. Root-Node Optimality Results

Figure EC.1 decomposes the aggregate results in Tables 1 and 2 of the main manuscript into instances proved optimal at the root node, additional instances proved optimal after branching, and instances that remain unsolved. Each bar is normalized by the corresponding dataset size: 150 for OPT and 1,200 for HEM. On OPT, BPC proves all 123 solved instances optimal before branching, whereas AFF proves 84 of its 104 solved instances optimal at the root node. On HEM, the corresponding root counts are 895 of 914 for BPC and 549 of 812 for AFF. These counts show that BPC owes most of its advantage to the number of instances proved optimal at the root node.

EC.4. Time-Window Length Sensitivity

Tables EC.2 and EC.3 report the number of instances proved optimal at each of the eight window lengths. Rows marked “Overall” aggregate the corresponding cost function or the complete data set.

Across all window lengths, Tables EC.2 and EC.3 reproduce the aggregate totals of 914/1,200 instances proved optimal by BPC and 812/1,200 by AFF. Both methods prove all 150 instances optimal at length 1. For each intermediate length from 2 to 64, BPC proves more instances optimal, with the largest difference at length 64 (94/150 versus 51/150). At length 128, time-window conflicts disappear and AFF proves 140/150 instances optimal, compared with 94/150 for BPC. Thus, the proposed framework is most beneficial when time windows induce nontrivial conflict structures.

EC.5. Performance by Conflict Density

In Tables EC.4 and EC.5, “Density” is the conflict probability d ; the other columns follow the definitions in the main manuscript and retain exactly the same meanings. Boldface marks the best or tied-best value among methods evaluated on the same instances. In count columns, only the count before the slash is bold and the denominator remains unbolded.

On SET-1, the BPC optimality count decreases from 188/240 at $d = 0$ to 143/240 at $d = 0.6$, then rises to 240/240 at $d = 0.99$; its average gap peaks at 0.1686% for $d = 0.7$.

On SET-2, BPC proves all 40 instances optimal for every cost function at $d = 0.99$. Together, these results indicate that low and intermediate conflict densities are generally the most difficult, whereas very dense graphs sufficiently restrict feasible patterns to simplify exact pricing and optimality proofs.

Table EC.2 Optimality Counts of BPC on HEM by Time-Window Length

Bin Cost	$ I $	#Opt by Window Length							
		1	2	4	8	16	32	64	128
Linear	100	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
	200	10/10	10/10	10/10	10/10	10/10	10/10	10/10	9/10
	500	10/10	10/10	10/10	9/10	10/10	6/10	5/10	5/10
	1000	10/10	10/10	9/10	9/10	4/10	4/10	6/10	6/10
	2000	10/10	6/10	1/10	3/10	5/10	4/10	4/10	5/10
	Overall	50/50	46/50	40/50	41/50	39/50	34/50	35/50	35/50
Concave	100	10/10	10/10	10/10	10/10	10/10	10/10	10/10	8/10
	200	10/10	10/10	10/10	10/10	10/10	8/10	3/10	6/10
	500	10/10	10/10	10/10	7/10	7/10	2/10	2/10	3/10
	1000	10/10	8/10	3/10	3/10	1/10	1/10	3/10	2/10
	2000	10/10	0/10	0/10	0/10	0/10	1/10	1/10	2/10
	Overall	50/50	38/50	33/50	30/50	28/50	22/50	19/50	21/50
Convex	100	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
	200	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
	500	10/10	10/10	10/10	10/10	10/10	10/10	9/10	7/10
	1000	10/10	10/10	10/10	10/10	4/10	3/10	6/10	6/10
	2000	10/10	10/10	10/10	9/10	4/10	5/10	5/10	5/10
	Overall	50/50	50/50	50/50	49/50	38/50	38/50	40/50	38/50
Overall		150/150	134/150	123/150	120/150	105/150	94/150	94/150	94/150

Table EC.3 Optimality Counts of AFF on HEM by Time-Window Length

Bin Cost	$ I $	#Opt by Window Length							
		1	2	4	8	16	32	64	128
Linear	100	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
	200	10/10	10/10	10/10	10/10	9/10	9/10	10/10	10/10
	500	10/10	10/10	10/10	8/10	8/10	1/10	0/10	10/10
	1000	10/10	10/10	6/10	5/10	0/10	0/10	0/10	10/10
	2000	10/10	4/10	1/10	0/10	0/10	0/10	0/10	10/10
	Overall	50/50	44/50	37/50	33/50	27/50	20/50	20/50	50/50
Concave	100	10/10	10/10	10/10	10/10	9/10	8/10	8/10	10/10
	200	10/10	10/10	10/10	8/10	5/10	6/10	4/10	8/10
	500	10/10	10/10	6/10	1/10	4/10	2/10	0/10	9/10
	1000	10/10	1/10	0/10	0/10	0/10	0/10	0/10	6/10
	2000	10/10	0/10	0/10	0/10	0/10	0/10	0/10	7/10
	Overall	50/50	31/50	26/50	19/50	18/50	16/50	12/50	40/50
Convex	100	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
	200	10/10	10/10	10/10	10/10	10/10	10/10	9/10	10/10
	500	10/10	10/10	10/10	10/10	10/10	8/10	0/10	10/10
	1000	10/10	10/10	10/10	9/10	3/10	0/10	0/10	10/10
	2000	10/10	10/10	9/10	1/10	0/10	0/10	0/10	10/10
	Overall	50/50	50/50	49/50	40/50	33/50	28/50	19/50	50/50
Overall		150/150	125/150	112/150	92/150	78/150	64/150	51/150	140/150

EC.6. Algorithmic Illustrations and Pseudocode

EC.6.1. Grouping Procedure and Label-Setting Algorithm Example

Figure EC.2 uses a five-item connected component of a compatibility graph that may occur within a larger root instance satisfying the GP activation conditions. Edges join item pairs without an original conflict. Panel (a) decomposes this component into its three maximal cliques, $Q_1 = \{1, 2, 4\}$, $Q_2 = \{2, 3, 4\}$, and $Q_3 = \{3, 4, 5\}$, which are also maximal cliques of the full compatibility graph and therefore define compatibility groups used by the GP. For the highlighted group Q_2 , consider a linear-cost bin type with $W_b = C_b = 10$, zero bin-type-bound duals, and no active 3-SRCs or RF branching relations. The item weights are $(w_2, w_3, w_4) =$

Table EC.4 Performance of LNSA and BPC on SET-1 by Conflict Density

Density	LNSA		BPC	
	Gap/%	#Opt	Gap/%	#Opt
0	0.09	117/240	0.0287	188/240
0.1	0.09	105/240	0.0207	182/240
0.2	0.10	91/240	0.0359	165/240
0.3	0.12	57/240	0.0547	152/240
0.4	0.15	27/240	0.0737	145/240
0.5	0.18	19/240	0.0947	152/240
0.6	0.28	20/240	0.1367	143/240
0.7	0.41	14/240	0.1686	155/240
0.8	0.68	9/240	0.1431	177/240
0.9	1.30	6/240	0.0269	211/240
0.95	1.89	5/240	0.0016	230/240
0.99	1.73	28/240	0.0000	240/240
Overall	0.58	498/2880	0.0654	2140/2880

Table EC.5 Performance of LNSA and BPC on SET-2 by Conflict Density

Convex					Linear				
Density	LNSA		BPC		Density	LNSA		BPC	
	Gap/%	#Opt	Gap/%	#Opt		Gap/%	#Opt	Gap/%	#Opt
0	5.67	0/40	0.00285	28/40	0	0.19	1/40	0.01684	25/40
0.1	5.66	0/40	0.00194	29/40	0.1	0.21	1/40	0.01869	23/40
0.2	5.65	0/40	0.00232	27/40	0.2	0.22	3/40	0.01477	24/40
0.3	5.66	0/40	0.00149	31/40	0.3	0.24	1/40	0.01843	27/40
0.4	5.65	0/40	0.00067	34/40	0.4	0.25	1/40	0.00924	30/40
0.5	5.61	0/40	0.00004	37/40	0.5	0.28	1/40	0.00176	36/40
0.6	5.69	0/40	0.00015	39/40	0.6	0.34	0/40	0.00484	31/40
0.7	6.14	0/40	0.00000	40/40	0.7	0.38	1/40	0.01796	37/40
0.8	5.98	0/40	0.00000	40/40	0.8	0.46	0/40	0.00040	39/40
0.9	5.74	0/40	0.00000	40/40	0.9	0.66	0/40	0.00000	40/40
0.95	5.20	0/40	0.00000	40/40	0.95	0.78	0/40	0.00000	40/40
0.99	3.28	0/40	0.00000	40/40	0.99	0.67	2/40	0.00000	40/40
Overall	5.50	0/480	0.00079	425/480	Overall	0.39	11/480	0.00858	392/480

Table EC.5 Performance of LNSA and BPC on SET-2 by Conflict Density (Continued)

Concave				
Density	LNSA		BPC	
	Gap/%	#Opt	Gap/%	#Opt
0	0.20	5/40	0.09721	10/40
0.1	0.22	6/40	0.40190	9/40
0.2	0.22	5/40	0.44066	10/40
0.3	0.23	3/40	0.24991	12/40
0.4	0.29	4/40	0.22912	9/40
0.5	0.26	4/40	0.15542	16/40
0.6	0.32	6/40	0.14747	16/40
0.7	0.44	3/40	0.10528	21/40
0.8	0.52	3/40	0.06315	27/40
0.9	0.85	1/40	0.00203	38/40
0.95	0.90	3/40	0.00000	40/40
0.99	0.54	12/40	0.00000	40/40
Overall	0.42	55/480	0.15768	248/480

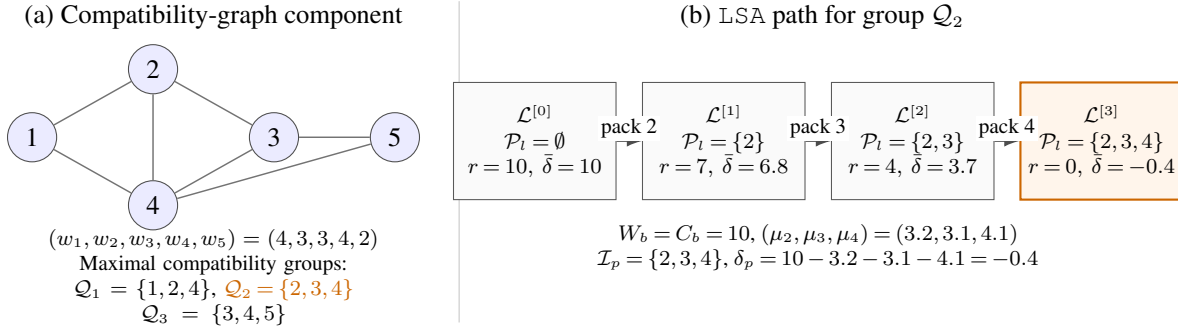


Figure EC.2 Grouping Procedure and Label-Setting Example

$(3, 3, 4)$ and the current item duals are $(\mu_2, \mu_3, \mu_4) = (3.2, 3.1, 4.1)$. Their dual-to-weight ratios give the processing order 2, 3, 4, and the highlighted packing extensions exactly fill the bin and produce a pattern p with $\mathcal{I}_p = \{2, 3, 4\}$ and $\delta_p = -0.4$. For $k \in \{0, 1, 2, 3\}$, $\mathcal{L}^{[k]}$ denotes the label on this path after the first k displayed packing extensions. The example represents a possible root pricing call before any 3-SRC or branching relation is active. The GP does not exclude a feasible pattern because every pairwise-compatible item set lies in at least one maximal clique of the compatibility graph. Panel (b) follows the packing successor at each LSA stage for the highlighted group. The LSA also creates the omitted nonpacking successors, applies its completion bounds and dominance tests, and returns the displayed pattern because its reduced cost is negative.

EC.6.2. Pattern Enumeration at the Root Node

Figure EC.3 summarizes the role of PE after the root CRG converges with a positive gap. Exact pricing has then established that the current RLMP dual solution is feasible for the full LMP, so all pattern reduced costs are nonnegative. Using the threshold $UB - z^R$, PE retains a finite pattern set containing every complete pattern that can participate in a solution cheaper than the incumbent. Each retained pattern defines a master column. Let z^E denote the optimal objective value of the resulting finite RSCP when it is feasible. Solving this model over the enumerated patterns and current columns can prove global optimality in two cases. Infeasibility under the cutoff $UB - 1$ proves the incumbent optimal, while an optimal feasible value z^E yields a new globally optimal incumbent. Reaching an enumeration or RSCP resource limit does not prove optimality and therefore returns control to the unrestricted BPC search.

For example, if $z^R = 100$ and $UB = 106$, PE retains every complete pattern with $\delta_p < 6$ and solves the RSCP with an integral cutoff of 105. If this finite model is infeasible, no packing has cost at most 105, so the incumbent value 106 is globally optimal, and the root node can be closed before unrestricted BPC search.

EC.6.3. Rounding Heuristic Example

Figure EC.4 illustrates the RH using five pair patterns that form an odd cycle. Such a solution can arise when capacity and conflicts permit these neighboring pairs. The five patterns form a common half-integral extreme point in which each pattern has $x_p = 0.5$, and every item has total LP coverage of one. The example scans the patterns, identifies the residual items, and uses the LCF heuristic to complete the packing.

All five patterns in the example use a bin type with $C_b = 6$. For an incumbent $UB = 24$, one possible order among equal LP values adds p_1 to $\mathcal{P}^c$, skips p_2 because item 2 is already packed, and adds p_3 , at which point $C(\mathcal{P}^c)$ reaches $\lfloor UB/2 \rfloor$. The LCF heuristic then packs $\mathcal{I}^u = \{5\}$ in pattern p_6 , producing the feasible integer solution $\mathcal{S}^{\text{RH}} = \{p_1, p_3, p_6\}$ with cost 18.

EC.6.4. Restricted Search in the Diving Heuristic

Figure EC.5 shows the restricted tree traversed by the DH. At each node, the left child is the binding branch and the right child is the conflict branch. A conflict branch consumes the single right-branch allowance on that root-to-node path, so subsequent conflict children on the same path are deferred. Filled nodes are processed during diving, whereas hollow nodes and their descendants remain available for the unrestricted search that follows if the instance is still unsolved.

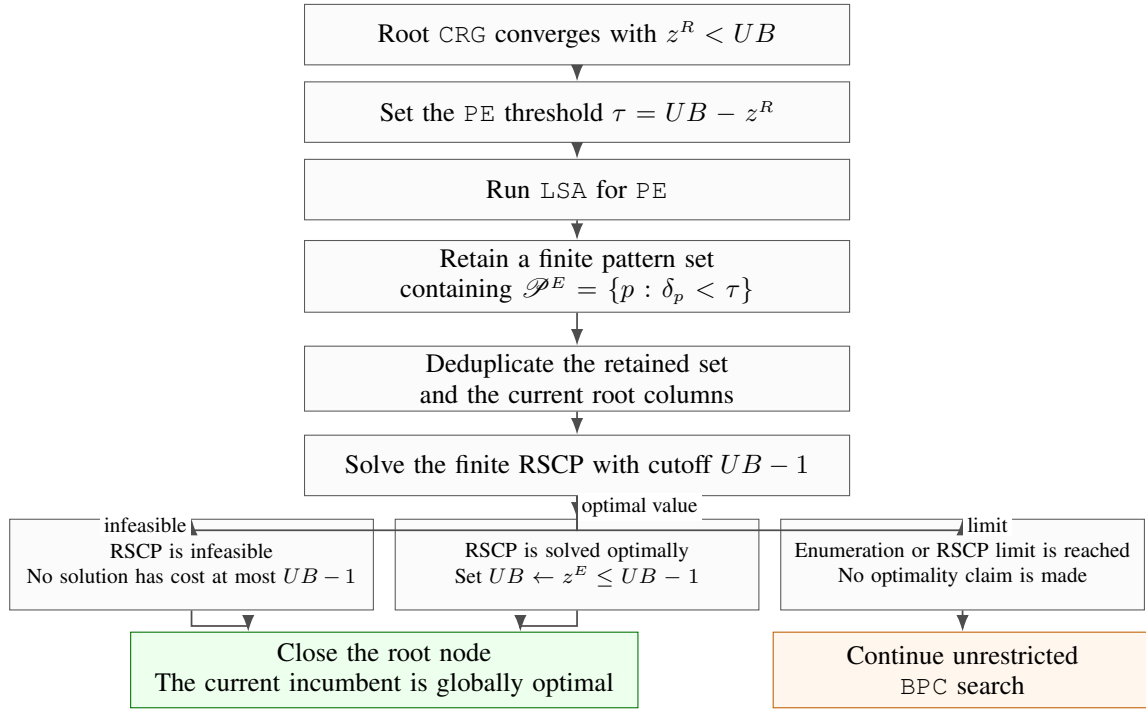


Figure EC.3 Pattern Enumeration at the Root Node

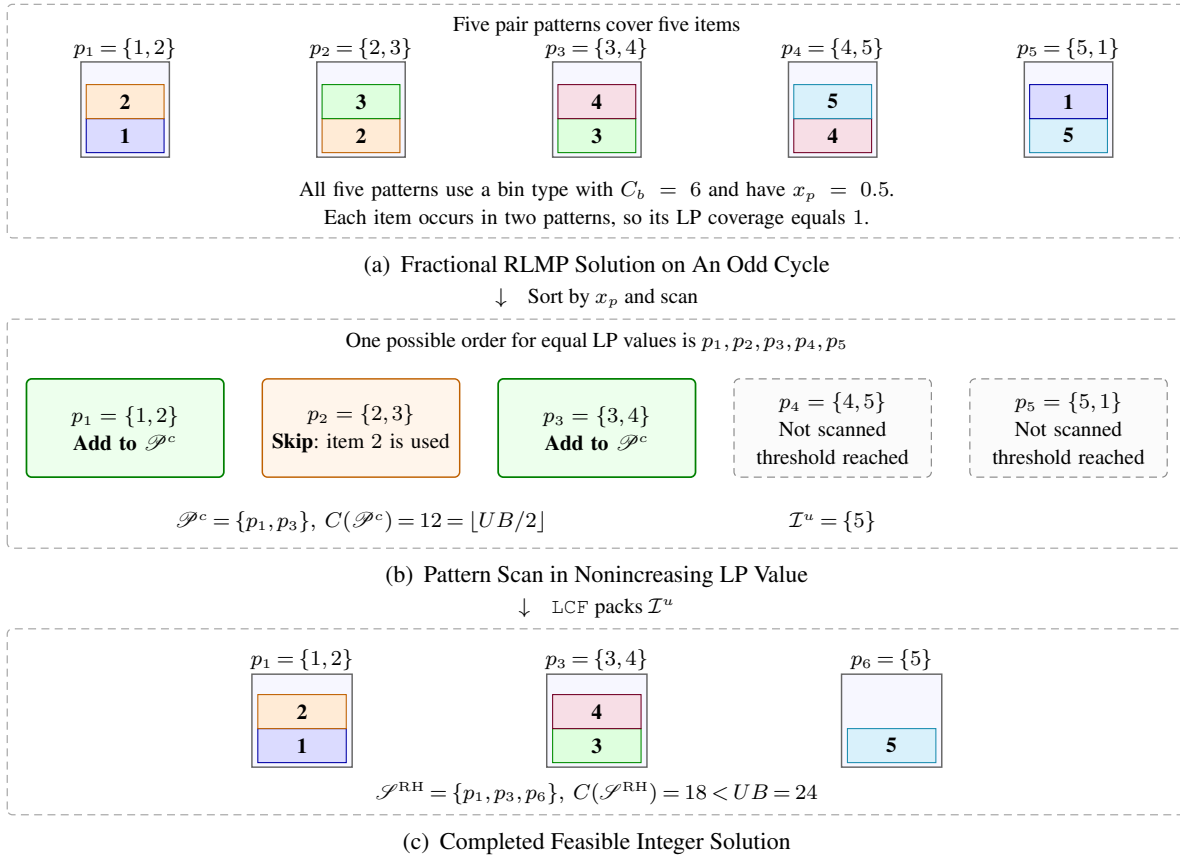


Figure EC.4 Rounding Heuristic Example

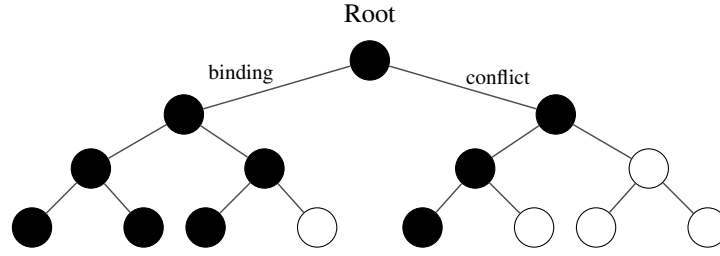


Figure EC.5 Restricted Search Tree of the Diving Heuristic

EC.6.5. Algorithm Pseudocode

Algorithm 1: Modified LCF Heuristic

Input: Item set $\mathcal{I}$ with weights w_i and conflict edge set $\mathcal{E}$; ordered bin-type set $\mathcal{B} = \{1, \dots, m\}$ with capacities $W_1 < \dots < W_m$ and costs C_b

Output: Feasible packing $\mathcal{S}^{\text{sol}}$ of typed patterns

- 1 Initialize $\mathcal{S}^{\text{sol}} \leftarrow \emptyset$ and the ordered set of unpacked items $\mathcal{A} \leftarrow \mathcal{I}$;
- 2 **foreach** $i \in \mathcal{A}$ **do**
- 3 Set $\nu_i \leftarrow |\{j \in \mathcal{A} \setminus \{i\} : w_i + w_j \leq W_m, \{i, j\} \notin \mathcal{E}\}|$;
- 4 **end**
- 5 Sort $\mathcal{A}$ by nonincreasing w_i , breaking ties by the smaller item index;
- 6 **while** $\mathcal{A} \neq \emptyset$ **do**
- 7 Select $i^* \in \mathcal{A}$ with the smallest ν_i , breaking ties by larger w_i and then by the smaller item index;
- 8 Initialize the selected pattern p^* as undefined and its score $\gamma^* \leftarrow +\infty$;
- 9 **for** $b = 1, \dots, m$ such that $w_{i^*} \leq W_b$ **do**
- 10 Initialize a type- b trial pattern $\hat{p}_b$ with $\mathcal{I}_{\hat{p}_b} \leftarrow \{i^*\}$;
- 11 **foreach** $j \in \mathcal{A} \setminus \{i^*\}$ in its current order **do**
- 12 **if** $\sum_{i \in \mathcal{I}_{\hat{p}_b}} w_i + w_j \leq W_b$ and $\{j, k\} \notin \mathcal{E}$ for all $k \in \mathcal{I}_{\hat{p}_b}$ **then**
- 13 Add j to $\mathcal{I}_{\hat{p}_b}$;
- 14 **end**
- 15 **end**
- 16 Set $\gamma_b \leftarrow C_b / \sum_{i \in \mathcal{I}_{\hat{p}_b}} w_i$;
- 17 **if** $\gamma_b \leq \gamma^*$ **then**
- 18 Set $p^* \leftarrow \hat{p}_b$ and $\gamma^* \leftarrow \gamma_b$;
- 19 **end**
- 20 **end**
- 21 Add the typed pattern p^* to $\mathcal{S}^{\text{sol}}$ and set $\mathcal{A} \leftarrow \mathcal{A} \setminus \mathcal{I}_{p^*}$ without changing the relative order of the remaining items;
- 22 **foreach** $i \in \mathcal{A}$ **do**
- 23 Update $\nu_i \leftarrow |\{j \in \mathcal{A} \setminus \{i\} : w_i + w_j \leq W_m, \{i, j\} \notin \mathcal{E}\}|$;
- 24 **end**
- 25 **end**
- 26 **return** $\mathcal{S}^{\text{sol}}$;

Algorithm 2: RH

Input: Current RLMP typed pattern set $\overline{\mathcal{P}} = \bigcup_{b \in \mathcal{B}} \overline{\mathcal{P}}_b$, where pattern p has item set $\mathcal{I}_p$ and LP value x_p ; item set $\mathcal{I}$ with weights w_i and conflict edge set $\mathcal{E}$; bin types $\mathcal{B}$; incumbent solution; global lower bound LB ; incumbent upper bound UB

Output: Updated incumbent solution and UB ; optimality status

- 1 Sort patterns in $\overline{\mathcal{P}}$ by nonincreasing x_p , retaining their current order when LP values are equal;
 - 2 Initialize the retained pattern set $\mathcal{P}^c \leftarrow \emptyset$ and its total cost $C(\mathcal{P}^c) \leftarrow 0$;
 - 3 **foreach** pattern $p \in \overline{\mathcal{P}}$ **in the resulting order** **do**
 - 4 **if** $\mathcal{I}_p \cap (\bigcup_{q \in \mathcal{P}^c} \mathcal{I}_q) = \emptyset$ **then**
 - 5 Add p to $\mathcal{P}^c$ and update $C(\mathcal{P}^c)$;
 - 6 **end**
 - 7 **if** $C(\mathcal{P}^c) \geq \lfloor UB/2 \rfloor$ **then**
 - 8 break;
 - 9 **end**
 - 10 **end**
 - 11 Set $\mathcal{I}^u \leftarrow \mathcal{I} \setminus \bigcup_{p \in \mathcal{P}^c} \mathcal{I}_p$;
 - 12 Apply Algorithm 1 to $\mathcal{I}^u$ with the conflicts induced by $\mathcal{E}$, and let $\mathcal{P}^u$ be the returned typed-pattern set;
 - 13 Set $\mathcal{P}^{\text{RH}} \leftarrow \mathcal{P}^c \cup \mathcal{P}^u$ and compute $C(\mathcal{P}^{\text{RH}})$;
 - 14 **if** $C(\mathcal{P}^{\text{RH}}) < UB$ **then**
 - 15 Update the incumbent solution to $\mathcal{P}^{\text{RH}}$ and set $UB \leftarrow C(\mathcal{P}^{\text{RH}})$;
 - 16 **end**
 - 17 Declare global optimality if $UB = LB$;
 - 18 **return** the incumbent solution, UB , and whether $UB = LB$;
-

Algorithm 3: LSA

Input: Ordered retained item set $\mathcal{I}^{\text{cur}} = \{1, \dots, n^{\text{cur}}\}$ with weights w_i , where $n^{\text{cur}} = |\mathcal{I}^{\text{cur}}|$, that is a union of complete current RF binding components and includes the prescribed zero-dual fallback when one exists; bin type b with capacity W_b and cost C_b ; item-covering duals μ_i ; lower and upper bin-type-bound dual values $\underline{\eta}_b$ and $\bar{\eta}_b$; active 3-SRC index set $\mathcal{T}$ with item triples $\mathcal{I}_t$ and duals ξ_t ; known pattern keys; active conflict and RF branching data

Output: Cutoff $\bar{\delta}^* = \min(\{0\} \cup \{\delta_p : p \text{ is a nonduplicate pattern first encountered in this call}\})$, where δ_p is the reduced cost of p ; record-improving pattern set $\mathcal{P}^{\text{rec}}$

- 1 Represent each label as $\mathcal{L} = (\mathcal{U}, \mathcal{H}, \bar{\delta}, r, \mathcal{P}_l, \bar{\delta}_{\min})$;
- 2 Here, $\mathcal{U}$ is the unavailable item set, $\mathcal{H} = \{t \in \mathcal{T} : |\mathcal{P}_l \cap \mathcal{I}_t| \text{ is odd}\}$ is the SR parity state, $\bar{\delta}$ is the current reduced cost, r is the residual capacity, $\mathcal{P}_l$ is the current partial pattern, and $\bar{\delta}_{\min}$ is the local lower bound;
- 3 Let $\mathcal{U}(\mathcal{L})$, $\mathcal{H}(\mathcal{L})$, $\bar{\delta}(\mathcal{L})$, $r(\mathcal{L})$, $\mathcal{P}_l(\mathcal{L})$, and $\bar{\delta}_{\min}(\mathcal{L})$ denote the corresponding components of label $\mathcal{L}$;
- 4 Define $\mathcal{C}(a)$ as the current RF binding component containing item a , and define $\mathcal{D}(\mathcal{L}, a)$ as the set of 3-SRCs newly activated when $\mathcal{C}(a)$ is added to $\mathcal{L}$;
- 5 Initialize $\mathcal{L}_0$ with $\mathcal{U} = \emptyset$, $\mathcal{H} = \emptyset$, $\bar{\delta} = C_b - \underline{\eta}_b - \bar{\eta}_b$, $r = W_b$, and $\mathcal{P}_l = \emptyset$, and compute $\bar{\delta}_{\min}(\mathcal{L}_0)$;
- 6 Initialize the stage label sets $\mathcal{L}_0 \leftarrow \{\mathcal{L}_0\}$ and $\mathcal{L}_a \leftarrow \emptyset$ for all $a = 1, \dots, n^{\text{cur}}$;
- 7 Initialize $\mathcal{P}^{\text{rec}} \leftarrow \emptyset$ and $\bar{\delta}^* \leftarrow 0$;
- 8 **for** $a = 1, \dots, n^{\text{cur}}$ **do**
- 9 Sort $\mathcal{L}_{a-1}$ by nondecreasing current reduced cost, nonincreasing residual capacity, and nondecreasing local lower bound;
- 10 Using only labels in $\mathcal{L}_{a-1}$ with nonempty partial patterns, delete each label whose local lower bound is no smaller than the smallest current reduced cost when such a cutoff exists; then apply the dominance rule in Proposition 2 to adjacent remaining labels only when the candidate dominating label has a nonempty partial pattern;
- 11 **foreach** $\mathcal{L} \in \mathcal{L}_{a-1}$ **do**
- 12 **if** $a \notin \mathcal{U}(\mathcal{L})$ and $\mathcal{C}(a)$ can be packed in $\mathcal{L}$ **then**
- 13 Create the packing successor $\mathcal{L}'$ with $\mathcal{P}_l(\mathcal{L}') \leftarrow \mathcal{P}_l(\mathcal{L}) \cup \mathcal{C}(a)$ and $r(\mathcal{L}') \leftarrow r(\mathcal{L}) - \sum_{i \in \mathcal{C}(a)} w_i$;
- 14 Set $\bar{\delta}(\mathcal{L}') \leftarrow \bar{\delta}(\mathcal{L}) - \sum_{i \in \mathcal{C}(a)} \mu_i - \sum_{t \in \mathcal{D}(\mathcal{L}, a)} \xi_t$ and update $\mathcal{H}(\mathcal{L}')$ by the parity rule;
- 15 Set $\mathcal{U}(\mathcal{L}')$ to include $\mathcal{U}(\mathcal{L})$, $\mathcal{C}(a)$, every retained item that conflicts with $\mathcal{C}(a)$, and the binding components of those items;
- 16 Compute $\bar{\delta}_{\min}(\mathcal{L}')$;
- 17 **if** $\bar{\delta}_{\min}(\mathcal{L}') < \bar{\delta}^*$ **then**
- 18 Add $\mathcal{L}'$ to $\mathcal{L}_a$;
- 19 **if** $\bar{\delta}(\mathcal{L}') < \bar{\delta}^*$ **then**
- 20 Convert $\mathcal{P}_l(\mathcal{L}')$ into a pattern p ;
- 21 **if** p is not a duplicate **then**
- 22 Update $\bar{\delta}^* \leftarrow \bar{\delta}(\mathcal{L}')$, add p to $\mathcal{P}^{\text{rec}}$, and record its key as known;
- 23 **end**
- 24 **end**
- 25 **end**
- 26 **end**
- 27 Create the nonpacking successor $\mathcal{L}''$ with $\mathcal{U}(\mathcal{L}'') \leftarrow \mathcal{U}(\mathcal{L}) \cup \mathcal{C}(a)$ and all other components unchanged;
- 28 Compute $\bar{\delta}_{\min}(\mathcal{L}'')$;
- 29 **if** $\bar{\delta}_{\min}(\mathcal{L}'') < \bar{\delta}^*$ **then**
- 30 Add $\mathcal{L}''$ to $\mathcal{L}_a$;
- 31 **end**
- 32 **end**
- 33 **end**
- 34 **return** $(\bar{\delta}^*, \mathcal{P}^{\text{rec}})$;

A full version of the EC is available at our GitHub repository: https://github.com/xiaopangZT/BPC_For_VSBPPC_TW

EC.7. Component Ablation Analysis

Tables EC.6 to EC.9 use the reporting conventions defined in the main manuscript. Each column group names the acceleration component disabled in that variant. Figure EC.6 summarizes the aggregate changes already reported in Table 5 of the main manuscript using the underlying unrounded averages. The left panel reports the reduction in optimality rate relative to the complete BPC, while the right panel reports the increase in average gap. “All off” denotes the variant without 3-SRCs, DH, PE, ET, and GP. Every tested removal increases the aggregate gap on both data sets, although the magnitude varies substantially across components.

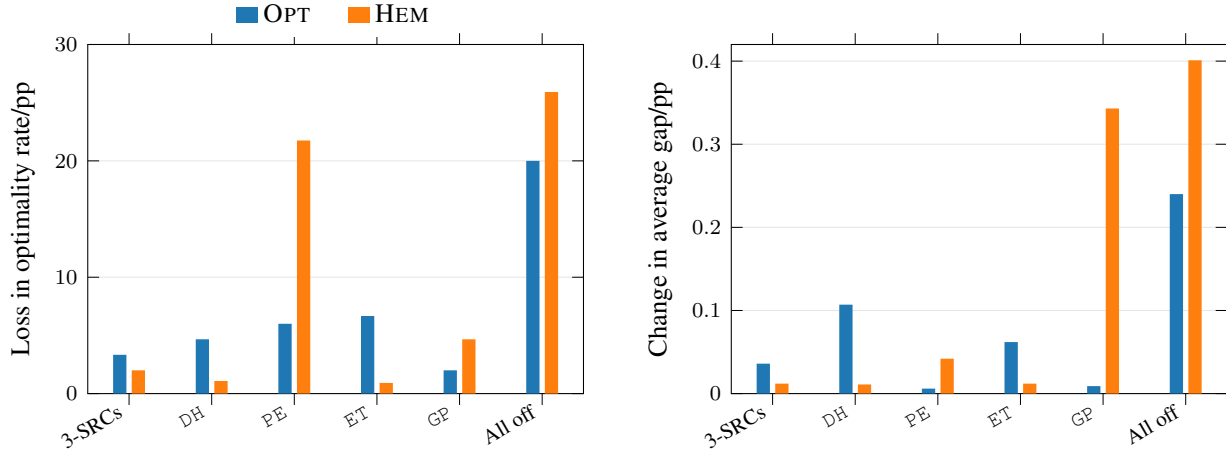


Figure EC.6 Aggregate Effects of Removing Acceleration Components

Figure EC.6 shows that disabling all five components causes the largest optimality-rate loss on both datasets. Among individual removals, PE reduces the HEM optimality rate the most, by 21.75 percentage points, while GP increases the average HEM gap the most, by 0.343 percentage points. On OPT, removing ET causes the largest optimality-rate decrease, and removing DH causes the largest gap increase. The metrics should be interpreted jointly because removing PE minimally affects the OPT gap but reduces its optimality rate more than several other removals.

Table EC.6 Ablation Results on OPT without 3-SRCs, DH, Or PE

Bin Cost	$ I $	No 3-SRCs			No DH			No PE		
		Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt
Linear	60	0.000	0.18	10/10	0.000	0.13	10/10	0.000	0.19	10/10
	120	0.000	0.46	10/10	0.000	0.47	10/10	0.000	0.39	10/10
	180	0.000	6.04	10/10	0.000	5.00	10/10	0.000	7.30	10/10
	240	0.000	85.84	10/10	0.000	80.03	10/10	0.025	465.62	5/10
	300	0.024	372.52	7/10	0.036	356.58	6/10	0.016	363.24	6/10
	Overall	0.005	93.01	47/50	0.007	88.44	46/50	0.008	167.35	41/50

Tables EC.6 and EC.7 disaggregate the OPT results underlying Figure EC.6. Most variants retain 10/10 instances proved optimal for sizes 60 and 120, while losses occur mainly at the larger sizes, particularly 240 and 300, and in the convex cost cases. Disabling all acceleration components reduces the aggregate count from 123/150 to 93/150. Among the individual variants, removing ET gives the lowest count, 113/150, followed by removing PE, with 114/150, whereas removing GP reduces the aggregate count from 123/150 to 120/150.

Table EC.8 shows substantially larger component effects on HEM. Removing PE reduces the aggregate count from 914/1,200 to 653/1,200, the largest individual loss among the variants reported in the table.

Table EC.6 Ablation Results on OPT without 3-SRCs, DH, Or PE (Continued)

Bin Cost	$ \mathcal{I} $	No 3-SRCs			No DH			No PE		
		Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt
Concave	60	0.000	0.06	10/10	0.000	0.06	10/10	0.000	0.07	10/10
	120	0.000	0.27	10/10	0.000	0.26	10/10	0.000	0.29	10/10
	180	0.000	2.30	10/10	0.000	0.77	10/10	0.000	0.79	10/10
	240	0.000	17.15	10/10	0.000	14.51	10/10	0.000	4.44	10/10
	300	0.254	361.84	5/10	0.415	434.53	3/10	0.357	481.56	2/10
	Overall	0.051	76.33	45/50	0.083	90.03	43/50	0.071	97.43	42/50
Convex	60	0.000	0.08	10/10	0.000	0.06	10/10	0.000	0.06	10/10
	120	0.000	6.56	10/10	0.000	1.20	10/10	0.000	0.91	10/10
	180	0.572	310.63	5/10	0.343	184.83	7/10	0.000	29.90	10/10
	240	0.775	541.20	1/10	1.544	600.01	0/10	0.775	588.55	1/10
	300	0.690	600.12	0/10	1.036	600.01	0/10	0.690	600.14	0/10
	Overall	0.408	291.72	26/50	0.585	277.22	27/50	0.293	243.91	31/50
Overall		0.154	153.68	118/150	0.225	151.90	116/150	0.124	169.56	114/150

Table EC.7 Ablation Results on OPT without ET, GP, Or All Acceleration Components

Bin Cost	$ \mathcal{I} $	No ET			No GP			No Acceleration Components		
		Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt
Linear	60	0.000	0.14	10/10	0.000	0.08	10/10	0.000	0.16	10/10
	120	0.000	0.49	10/10	0.000	0.52	10/10	0.000	0.46	10/10
	180	0.000	4.56	10/10	0.000	5.01	10/10	0.000	18.80	10/10
	240	0.000	100.12	10/10	0.000	76.99	10/10	0.060	475.13	3/10
	300	0.068	489.17	2/10	0.036	351.44	6/10	0.076	542.25	1/10
	Overall	0.014	118.90	42/50	0.007	86.81	46/50	0.027	207.36	34/50
Concave	60	0.000	0.07	10/10	0.000	0.07	10/10	0.000	0.07	10/10
	120	0.000	0.27	10/10	0.000	0.31	10/10	0.000	0.30	10/10
	180	0.000	0.85	10/10	0.000	1.20	10/10	0.000	1.33	10/10
	240	0.000	13.75	10/10	0.000	13.34	10/10	0.322	301.71	5/10
	300	0.578	439.73	3/10	0.408	434.74	3/10	0.640	540.54	1/10
	Overall	0.116	90.93	43/50	0.082	89.93	43/50	0.192	168.79	36/50
Convex	60	0.000	0.06	10/10	0.000	0.05	10/10	0.000	0.07	10/10
	120	0.000	1.24	10/10	0.000	1.26	10/10	0.000	3.20	10/10
	180	0.229	168.87	8/10	0.000	30.99	10/10	1.062	421.51	3/10
	240	0.861	600.02	0/10	0.775	587.48	1/10	1.785	600.01	0/10
	300	0.975	600.04	0/10	0.690	600.18	0/10	1.434	600.06	0/10
	Overall	0.413	274.05	28/50	0.293	243.99	31/50	0.856	324.97	23/50
Overall		0.181	161.29	113/150	0.127	140.25	120/150	0.359	233.71	93/150

Table EC.9 shows that removing GP produces the largest individual increase in average gap, from 0.016% to 0.359%. Disabling all five components gives the weakest aggregate result, with 603/1,200 instances proved optimal and an average gap of 0.418%. Across the three cost functions, deterioration concentrates in larger instances, where pricing and enumeration benefit most from the acceleration mechanisms.

Table EC.8 Ablation Results on HEM without 3-SRCs, DH, Or PE

Bin Cost	$ I $	No 3-SRCs			No DH			No PE		
		Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt
Linear	100	0.002	8.52	79/80	0.000	0.98	80/80	0.003	24.90	77/80
	200	0.001	8.27	79/80	0.001	8.13	79/80	0.003	37.30	76/80
	500	0.004	126.75	65/80	0.004	124.92	65/80	0.018	320.03	39/80
	1000	0.004	226.63	54/80	0.004	219.74	55/80	0.024	442.94	23/80
	2000	0.012	370.73	38/80	0.012	355.69	38/80	0.030	467.07	20/80
	Overall	0.005	148.18	315/400	0.004	141.89	317/400	0.016	258.45	235/400
Concave	100	0.013	16.46	78/80	0.018	24.09	77/80	0.068	132.65	63/80
	200	0.045	114.27	65/80	0.038	102.96	67/80	0.097	263.68	47/80
	500	0.026	259.19	47/80	0.024	257.81	48/80	0.108	504.10	15/80
	1000	0.200	426.47	27/80	0.195	420.99	28/80	0.302	567.27	6/80
	2000	0.120	508.49	14/80	0.114	510.07	14/80	0.209	577.95	4/80
	Overall	0.081	264.98	231/400	0.078	263.18	234/400	0.157	409.13	135/400
Convex	100	0.000	0.02	80/80	0.000	0.02	80/80	0.000	3.92	80/80
	200	0.000	0.34	80/80	0.000	0.28	80/80	0.001	52.26	74/80
	500	0.001	47.14	74/80	0.001	46.73	74/80	0.001	64.93	72/80
	1000	0.002	189.91	57/80	0.002	174.61	58/80	0.003	350.79	36/80
	2000	0.002	262.10	53/80	0.001	219.51	58/80	0.002	482.86	21/80
	Overall	0.001	99.90	344/400	0.001	88.23	350/400	0.001	190.95	283/400
Overall		0.029	171.02	890/1200	0.028	164.44	901/1200	0.058	286.18	653/1200

Table EC.9 Ablation Results on HEM without ET, GP, Or All Acceleration Components

Bin Cost	$ I $	No ET			No GP			No Acceleration Components		
		Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt	Gap/%	Time/s	#Opt
Linear	100	0.000	1.15	80/80	0.000	0.91	80/80	0.006	31.60	76/80
	200	0.001	8.06	79/80	0.001	8.13	79/80	0.005	54.49	75/80
	500	0.003	106.55	67/80	0.004	124.66	65/80	0.064	428.24	24/80
	1000	0.004	219.11	56/80	0.003	238.46	58/80	0.055	485.79	17/80
	2000	0.012	357.26	39/80	1.190	540.74	12/80	1.190	541.17	12/80
	Overall	0.004	138.42	321/400	0.240	182.58	294/400	0.264	308.26	204/400
Concave	100	0.018	24.04	77/80	0.018	24.09	77/80	0.088	149.31	61/80
	200	0.038	100.06	67/80	0.038	103.05	67/80	0.197	328.40	37/80
	500	0.024	254.13	48/80	0.024	259.53	48/80	0.308	551.15	7/80
	1000	0.200	435.51	26/80	0.257	429.90	28/80	0.469	575.69	5/80
	2000	0.126	510.53	14/80	2.293	585.78	3/80	2.309	586.80	3/80
	Overall	0.081	264.85	232/400	0.526	280.47	223/400	0.674	438.27	113/400
Convex	100	0.000	0.02	80/80	0.000	0.02	80/80	0.000	6.32	80/80
	200	0.000	0.26	80/80	0.000	0.28	80/80	0.001	50.45	74/80
	500	0.001	46.62	75/80	0.001	46.92	74/80	0.001	72.51	73/80
	1000	0.002	187.33	57/80	0.002	182.39	58/80	0.007	335.62	42/80
	2000	0.001	219.91	58/80	1.556	374.19	49/80	1.564	535.89	17/80
	Overall	0.001	90.83	350/400	0.312	120.76	341/400	0.315	200.16	286/400
Overall		0.029	164.70	903/1200	0.359	194.60	858/1200	0.418	315.56	603/1200